A matrix can be replaced by *two thin ones.*

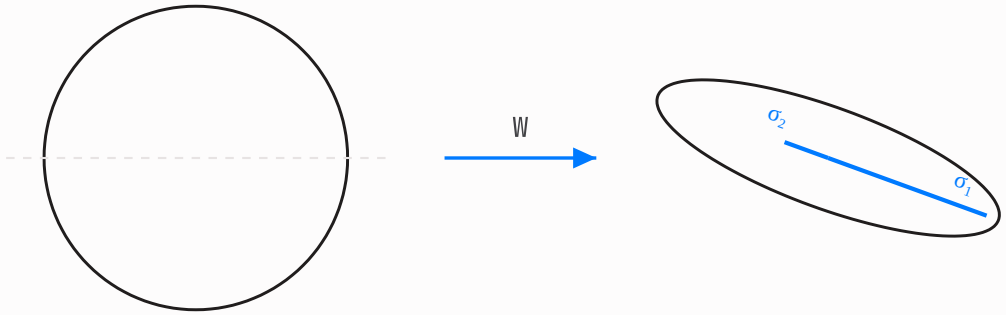
Eckart–Young says how much you lose. It does not say what you lose it for — and the difference between those two questions is where a neural network actually gets compressed.

A weight matrix is a shape

A matrix $W \in \mathbb{R}^{m \times n}$ is a linear map, and the honest picture is geometric: it sends the unit sphere to an ellipsoid.

The semi-axes of that ellipsoid are the **singular values** $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$. They say how much the map does in each direction — and the natural hope is that most of them say *almost nothing*.

geometry only · no learning yet



input · every direction equal

output · an ellipsoid

[+] THE DECOMPOSITION

Rotate, stretch, rotate

$$\mathbb{W} = \mathrm{U}\,\Sigma\,\mathrm{V}^{\mathrm{T}}, \quad \Sigma = \mathrm{diag}(\sigma_1 \geq \sigma_2 \geq \cdots \geq 0)$$

FACTOR	WHAT IT DOES
V^T	turns the input onto the principal axes
Σ	stretches axis *i* by σ_i
U	turns the result into output space

The σ_i say *how much* the map does in each direction. If the tail of the spectrum is negligible, the matrix is, for practical purposes, a much smaller matrix.

[+] THE THEOREM

Eckart–Young, 1936

Let $W_r = \Sigma_{i \le r}\ \sigma_i\ u_i\ v_i^{\top}$. Then for *every* matrix B with rank B ≤ r,

$$\|W - W_r\|_F \le \|W - B\|_F, \quad \|W - W_r\|_F = \sqrt{(\Sigma_{i>r}\ \sigma_i^2)}$$

01 the truncated SVD is the best rank-r approximation — optimal, not heuristic

02 its error is known in advance: the tail of the spectrum

03 no assumption on the data, the training, or the matrix being nice

the strongest statement here — and the one most often misread

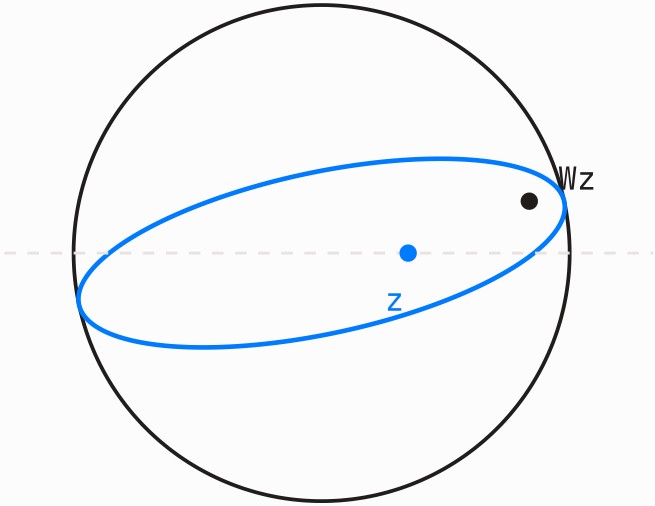
No subspace does better

Any rank-r candidate B has a null space of dimension ≥ n - r, so it contains a unit vector z with **Bz = 0**. On that z:

$$\|W - B\|_F \geq \|(W - B)z\| = \|Wz\|$$

Choosing z inside span(v_{r+1}, …, v_n) gives $\|Wz\| \geq \sigma_{r+1}$, and summing over the remaining directions reproduces exactly $\sqrt{(\sigma_{r+1}^2 + \cdots)}$.

no optimisation · no training · true for every matrix

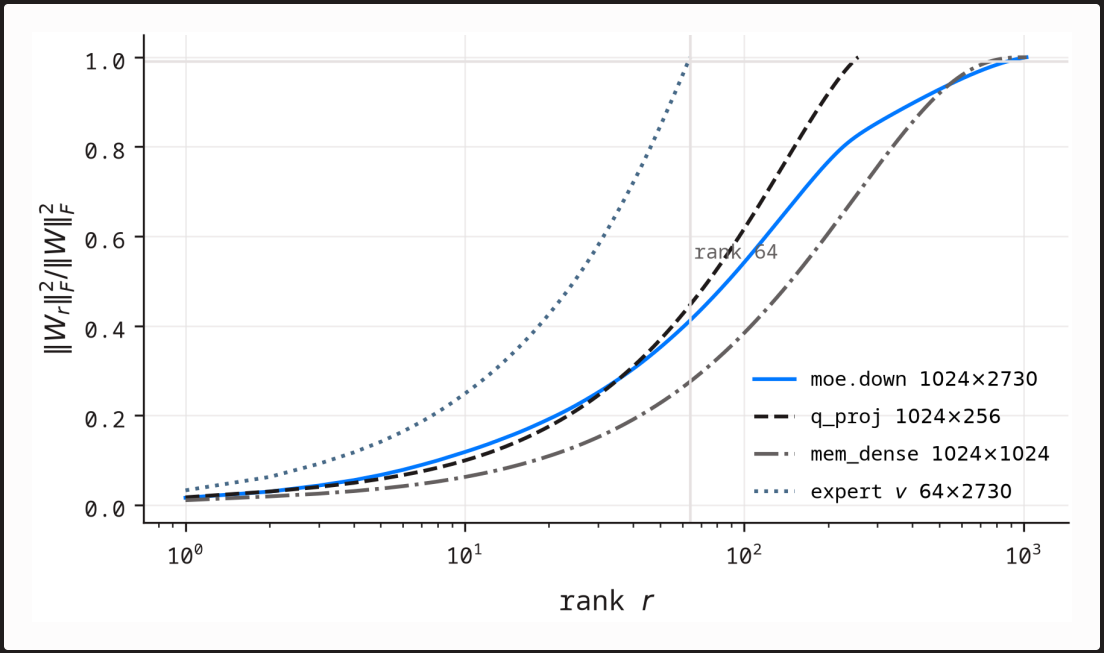


energy the candidate cannot reproduce

What do trained matrices look like?

[+] THE SURPRISE

The expectation fails



MATRIX	MASS AT R = 64
moe.down	41%
q_proj	45%
mem_dense	28%
expert v	100%

Trained matrices are not nearly low-rank.

Truncating a trained layer to rank 64 throws away more than half of it by the theorem's own measure. The last row is different for a structural reason — next slide.

[+] THE RESOLUTION

Compression ≠ reparameterisation

The expert matrix is rank 64 by construction: we never store a 64×2730 matrix — we store two thin ones whose product has rank ≤ 64.

$$\mathbb{W} = \mathrm{U}\,\mathrm{V}^{\mathrm{T}},\quad \mathrm{U} \in \mathbb{R}^{64\times 64},\quad \mathrm{V} \in \mathbb{R}^{64\times 2730}$$

compression

Given a fixed W, how well can rank r imitate it?

answer: badly

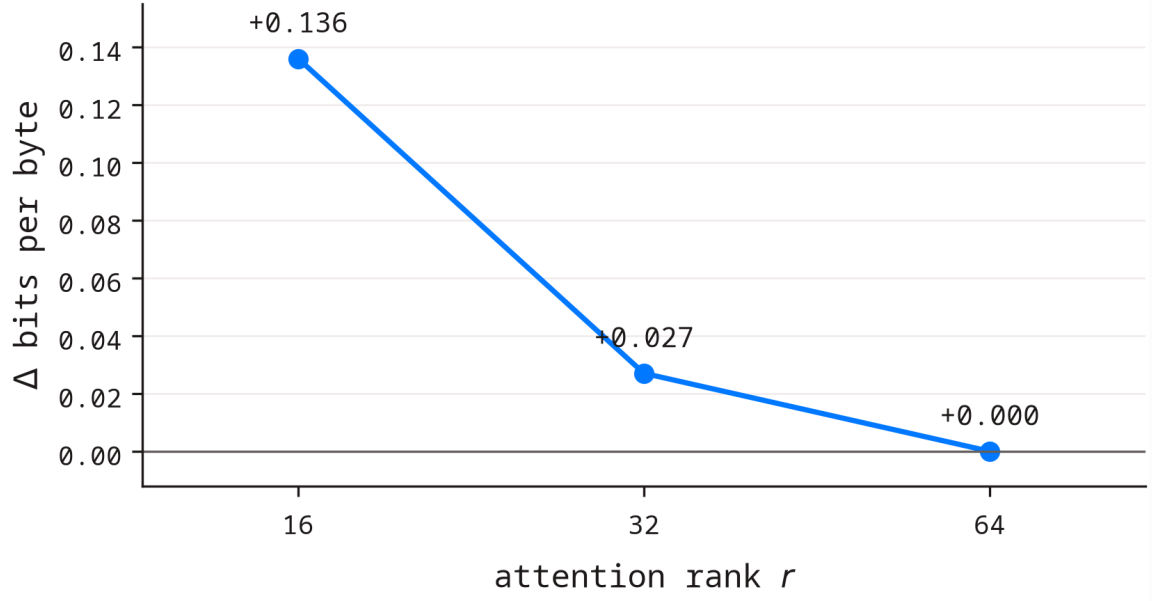
reparameterisation

Restrict the layer to rank r *before* training, and let descent search there.

answer: almost free

Eckart-Young bounds the first. It says nothing about the second — and the second is what gets deployed.

Measured, not assumed



three seeds per arm · identical budget · the same data

RANK	Δ BITS/BYTE	STORAGE
64	baseline	1.00×
32	+0.027	0.50×
16	+0.136	0.25×

Halving the rank costs about 2% of quality; quartering it costs 10%.

The spectrum predicts the *shape* of this curve. Only the experiment gives its height.

[+] THREE THINGS TO TAKE AWAY

A shape, a surprise, and a distinction

01 **a matrix is a shape** — the singular values are its semi-axes, and the truncated SVD is the optimal rank-r replacement, with the error known in advance.

02 **trained matrices are not low-rank** — measured: rank 64 keeps 28–45% of the mass. The naive compression story is false for real weights.

03 **compress ≠ reparameterise** — truncating is expensive, training inside the family is nearly free, and only the first is covered by the theorem.

exercise — take any layer, compute σ_i, predict the loss after truncation with $\sqrt{(\Sigma_{i>r}\ \sigma_i{}^2)}$, then measure.
the ratio between prediction and reality is the interesting number.